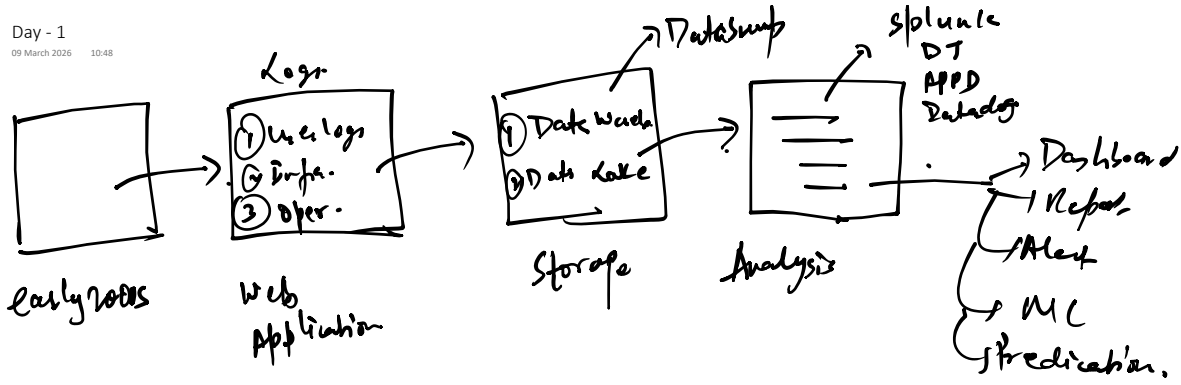




Monitoring Vs

① Reactive Approach.

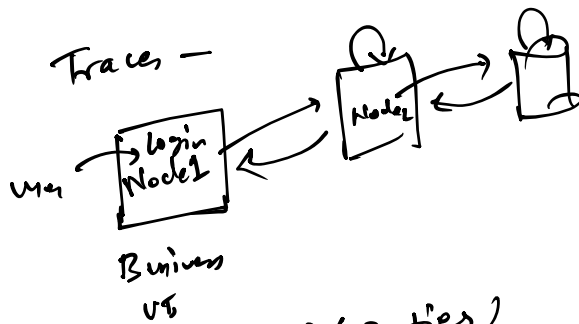
② Log Monitoring

Observability
① Proactive Approach.

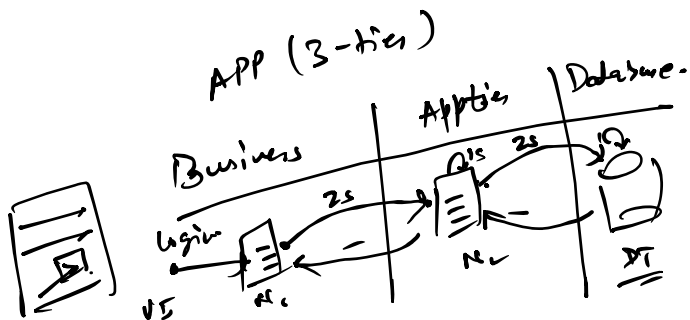
② Logs, Traces, Metrics. → Detail over here

Metrics - CPU = 80%

Logs - Timestamp Value -
-time

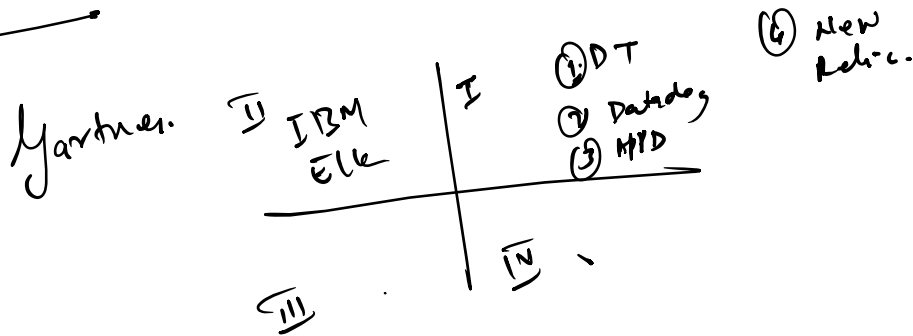


APP (2-tier)
Business | App tier.

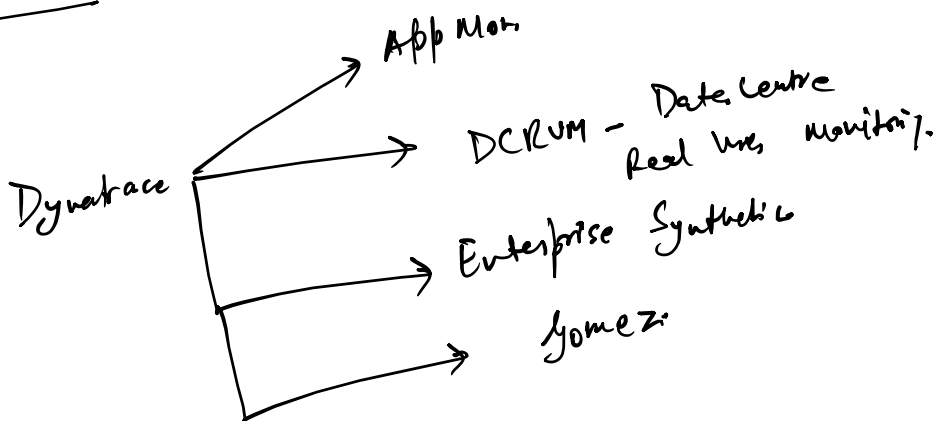


- ① Log Monitoring. → Logs coming from different sources.
- ② Application Performance Monitoring
- ③ End user monitoring (EUM) → Real User Monitoring
- ④ Container Monitoring → Containers. → Synthetic Monitoring.
- ⑤ Cloud Monitoring. → AWS, Azure, GCP...
- ⑥ Database Visibility. → MySQL, MongoDB, Oracle, MS SQL...

Dynatrace :- APM, AIOps, Cloud Infra., DEM

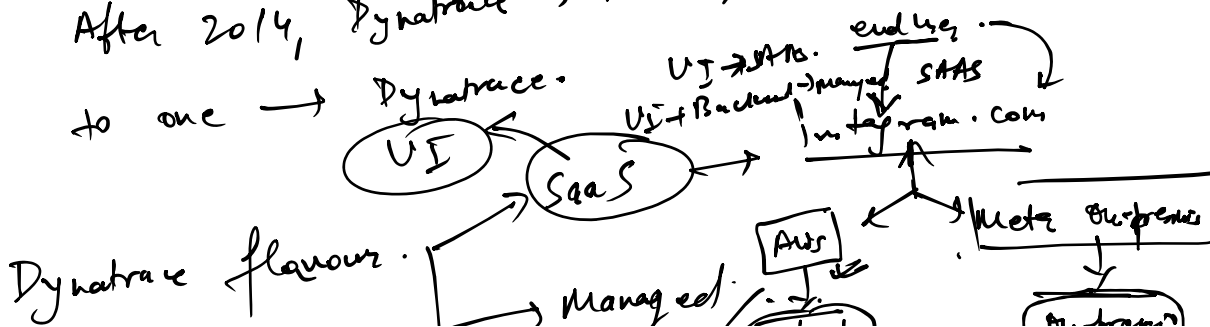


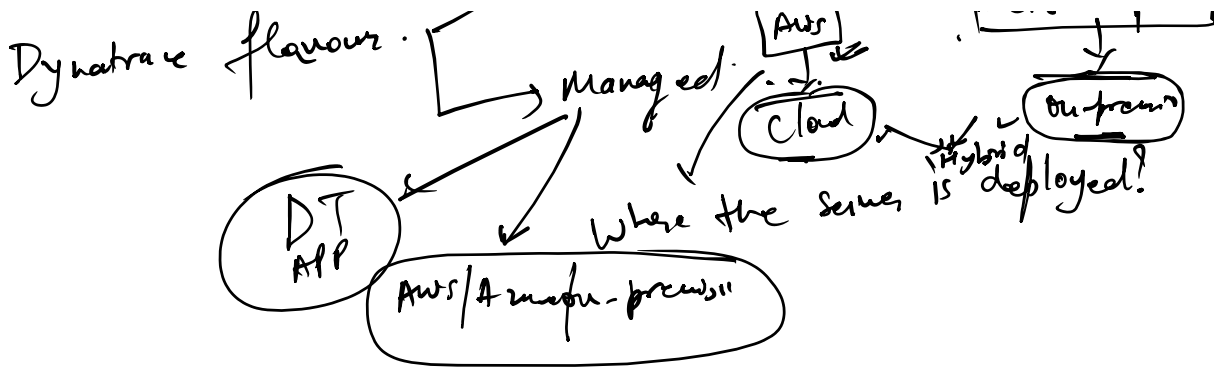
Before 2014 :-



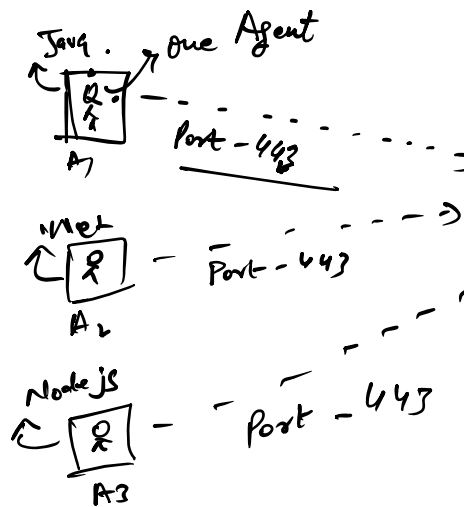
After 2014, Dynatrace → Ruxit, Combine all 4 app

to one → Dynatrace.





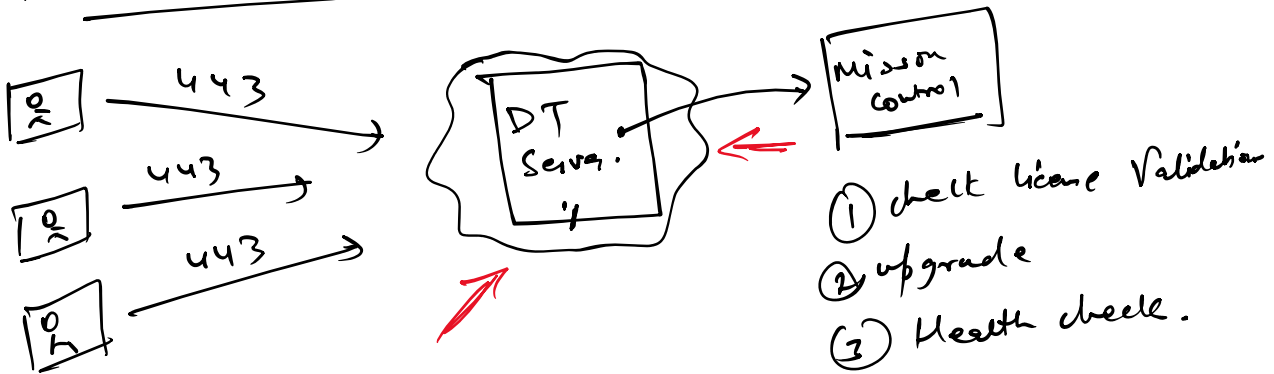
① DT SAAS :-



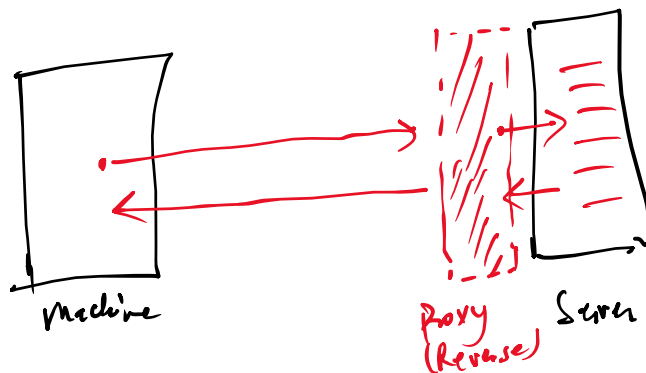
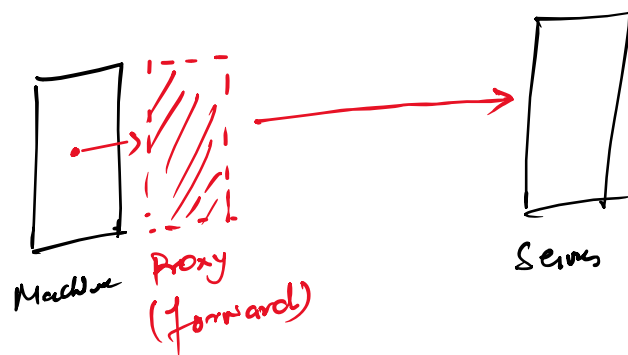
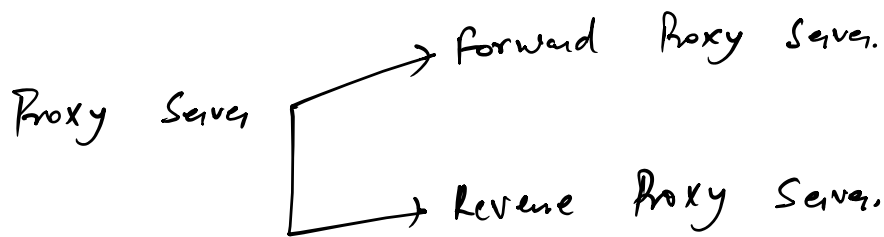
One Agent,

- ① Do n't depend on App type.
- ② Dependency under big OS of the platform Windows, Linux etc.

② Mission Control :- It will run on Dynatrace.



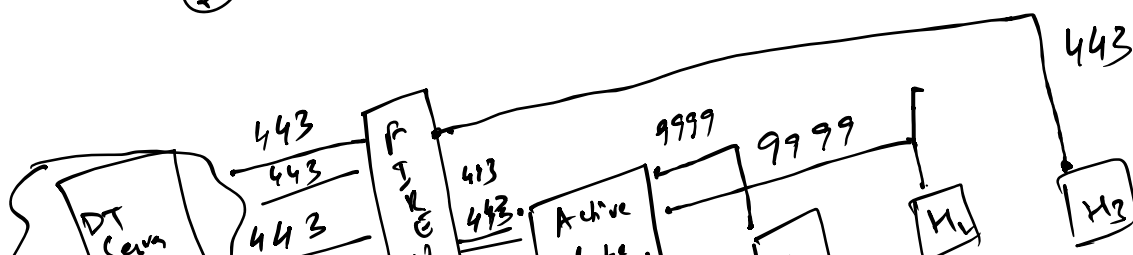
- ① Ingress → Reverse Proxy Server.
- ② Elastic Search → Search engine.
- ③ Cassandra Hypercube → Distributed Database where data is stored.
- ④ DT Server → Collects, Process & Analyse the data.

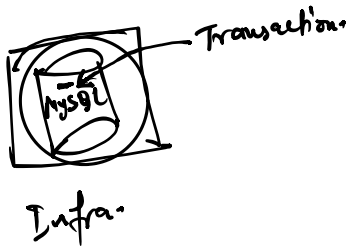
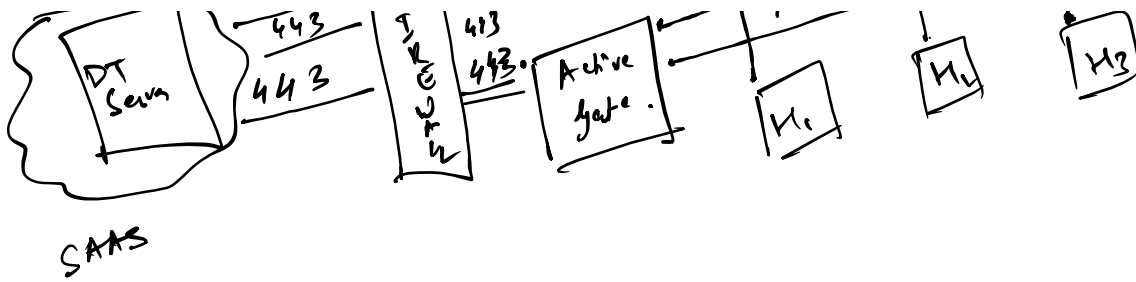


Environment Active Gate (AG):- Additional Component that only comes with SaaS environment.

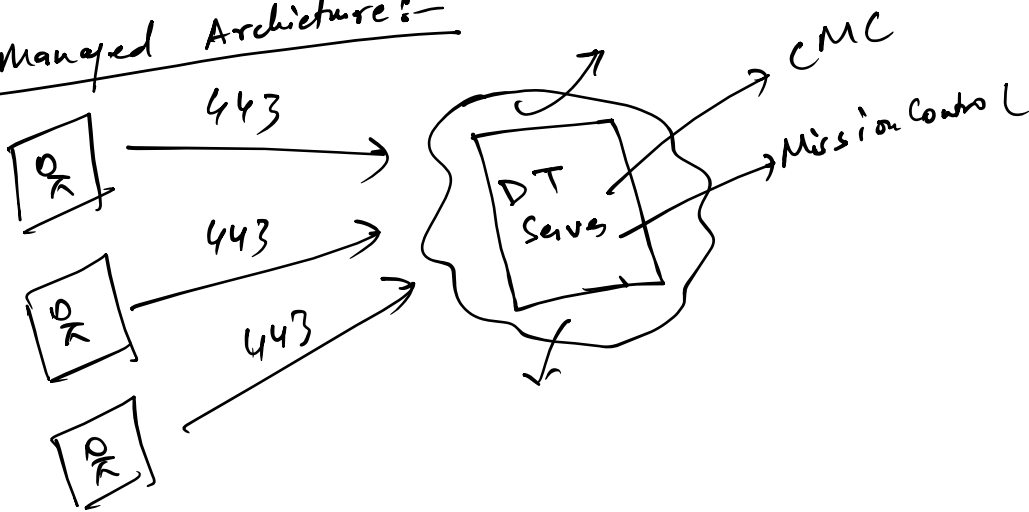
What? why?

- ① Cloud Integration.
- ② Database Visibility
- ③ Access the private URL.





DT Managed Architecture:-



H/W specification:-

Micro - 50 nodes - 4 CPU, 32GB RAM

Small - 300 nodes - 8 CPU, 64GB

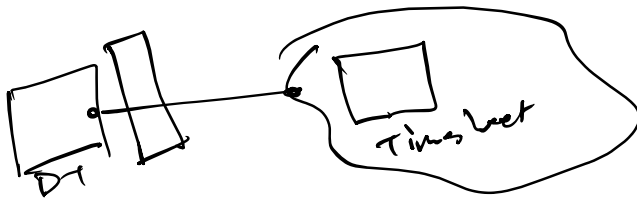
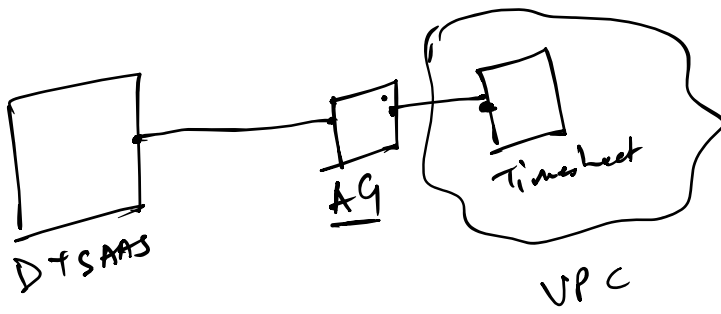
Medium - 600 nodes - 16 CPU, 128GB

Large - 1250 nodes - 32 CPU, 256GB RAM

X Large - 2500 nodes - 64 CPU, 512GB RAM

Additional Component - SAAS - Active Gate

Managed x



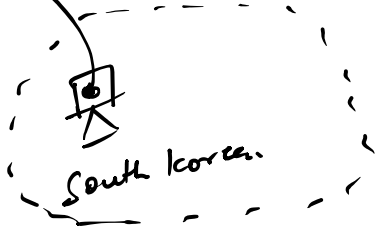
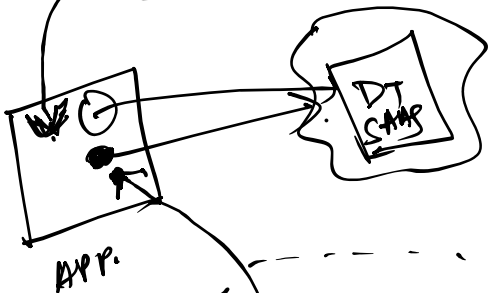
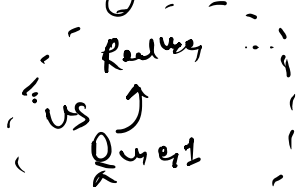
Discovery → SmartScope, Host, Basic Monitoring → Foundation.

Infrastructure → SmartScope, Host, Basic Monitoring, Host Access, Disk Analysis

Full Stack
 ↳ SmartScope, Host, Basic Monitoring, Host Access, Disk Analysis, Application tracing, security
User experience

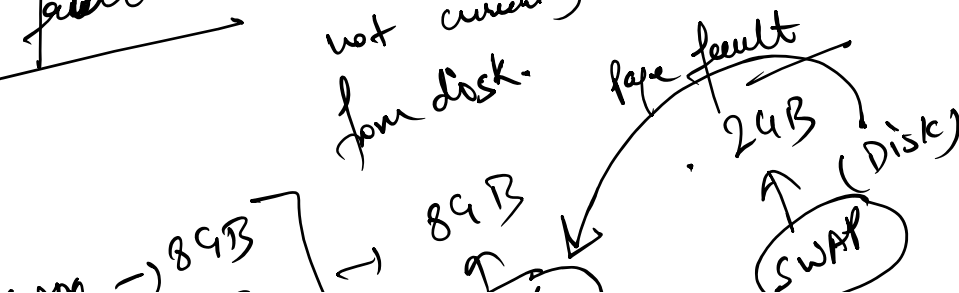


Dynatrace

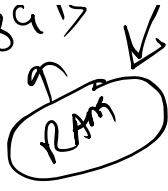


Page fault :-

Happens when a CPU needs data that is not currently in RAM, so the OS loads it from disk.



RAM $\rightarrow$ 8GB
 App $\rightarrow$ 10GB



SWAP

Page fault $\uparrow$
 RAM $\uparrow$

System become slow
 App. lag
 } Memory Pressure.

Swap Usage:-

Back memory on disk.

8GB $\rightarrow$ RAM
 10GB $\rightarrow$ App.

Remaining $\rightarrow$ 2GB moved to swap

$\downarrow$
 used from disk
 instead of RAM.

SWAP $\uparrow$ usage of disk $\uparrow$ System Performance $\downarrow$

XHR

$\rightarrow$ XMLHttpRequest Request

Web page sends the req. to the server
 in the background w/o reloading the page

Asynchronous web App.

User Action

$\downarrow$
 Browser send XHR req.

$\downarrow$
 Server process req. $\rightarrow$ Server send response $\rightarrow$ Page updates Dynamically.

Interaction to the Next Paint (INP):-

Web Performance Metric that measure how quickly a website respond when a user interact with it.

user click button $\rightarrow 0ms$
Browser process Action $\rightarrow 120ms$
Page updates (next paint) $\rightarrow 180ms$

INP = 200ms

INP $\uparrow$ website slow or loggy

INP $< 200ms$ $\rightarrow$ good (fast response)

INP = (200 - 500ms) $\rightarrow$ need improvement

INP $> 500ms$ $\rightarrow$ poor (Loggy UI)

High INP:-

Common Reason:-

Heavy JS
slow API calls.
Large DOM updates

CPU overload in browser.

Errors:-

- $\rightarrow$ Request Error (HTTP resp. code) (400-500)
 $\uparrow$ error.
- $\rightarrow$ Javascript Error (js) - error.
- or. (specific error after you
trace)

